

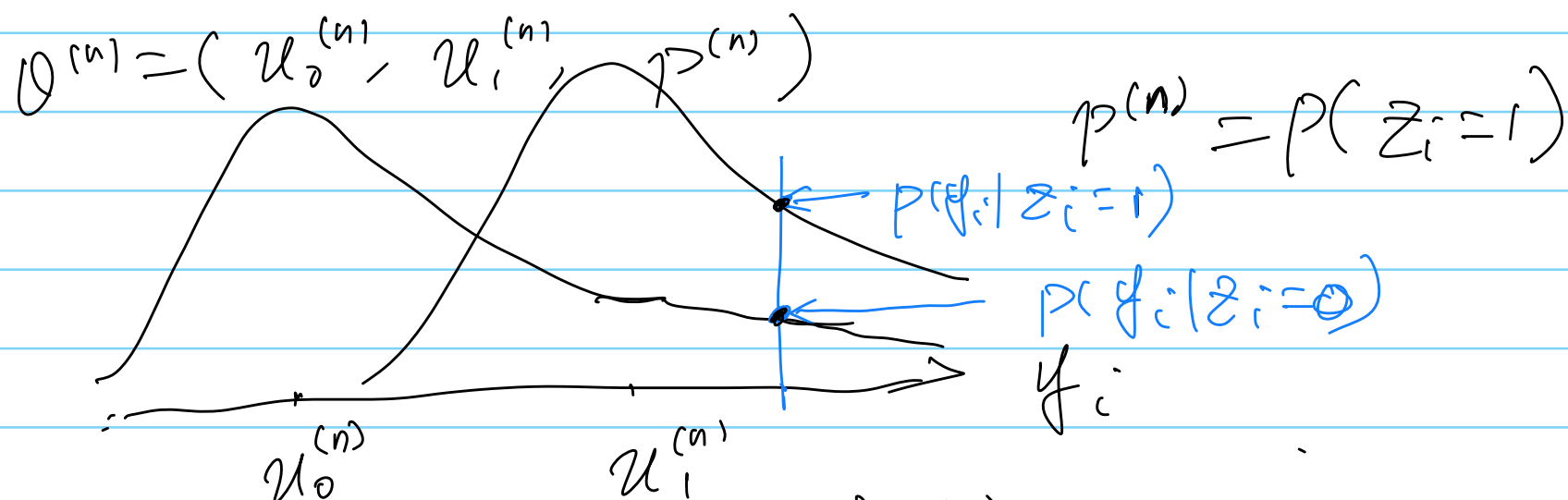
Lecture Notes for Computational Statistics

EM Algorithm

- Fitting Mixture Models
- Fitting Censored Data Models

Longhai Li
Department of Mathematics and Statistics
University of Saskatchewan

E-Step: finding soft classification probabilities $p_i^{(n)}$ for fitting mixture distribution with EM

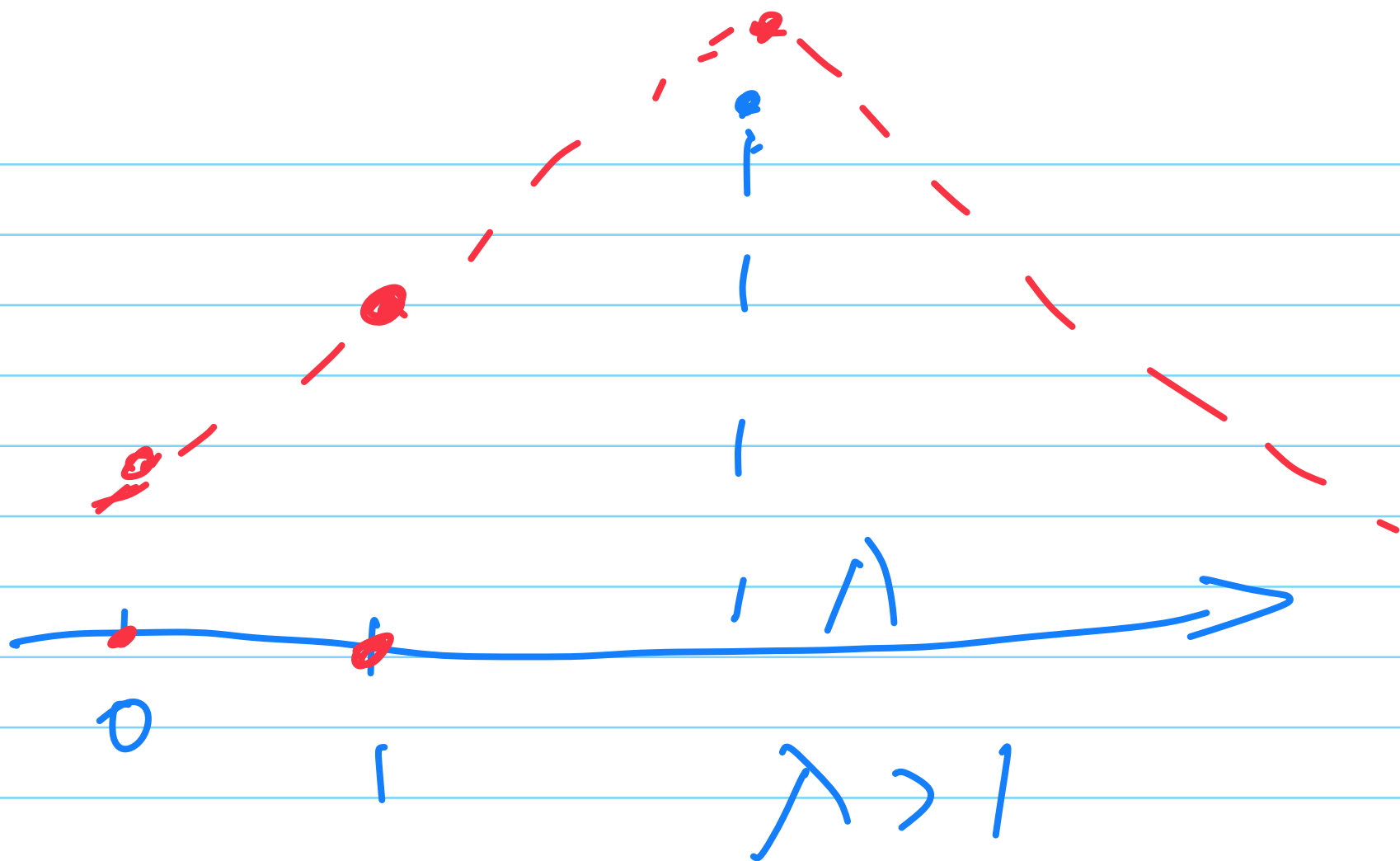


$$P(z_i = 1 | y_i, \theta^{(n)}) = \frac{P(y_i | z_i = 1) p^{(n)}}{P(y_i | z_i = 0) \cdot (1 - p^{(n)}) + P(y_i | z_i = 1) \cdot p^{(n)}}$$

$$\equiv p_i^{(n)} \text{ (soft classification)}$$

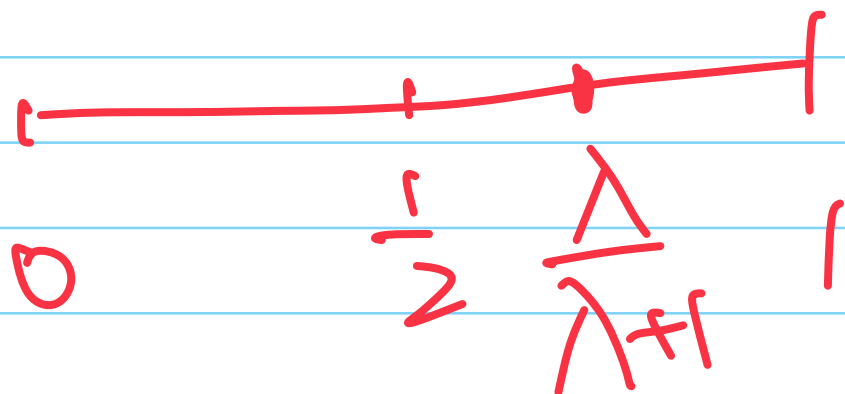
y_i	$\hat{p}_i^{(n)}$	$P(y_i z_i = 0) \cdot (1 - p^{(n)})$	$P(y_i z_i = 1) \cdot p^{(n)}$
x	$\frac{25}{45}$	20	25
x	0.2	x	x
x	0.9	,	,
x	.	,	,
x	.	,	,
x	.	,	,
,	,	,	,
,	,	,	,
,	,	,	,

$$\hat{p}^{(n+1)} = \frac{\sum \hat{p}_i^{(n)}}{N}$$



$$P(Y=1 \mid Y < 2) \Rightarrow \frac{1}{2}$$

$$E(Y \mid Y < 2) = \frac{\lambda}{1+\lambda}$$



$$\rho < 1$$



$$P(y=1 | y < 2) < \frac{1}{2}$$

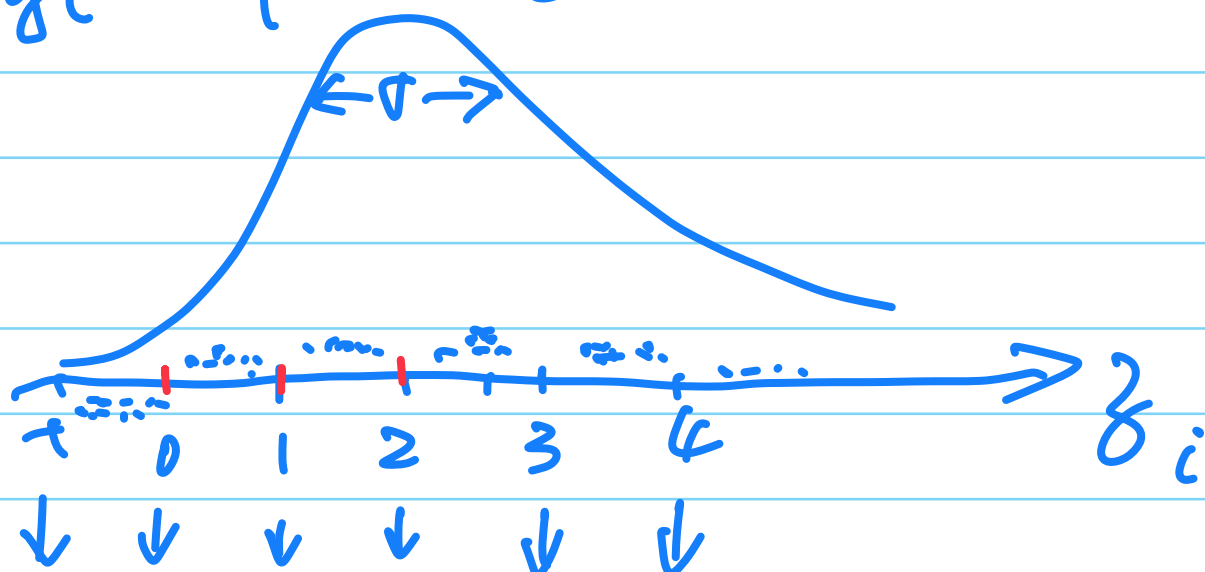
$$E(y | y < 2) = \frac{\Delta}{(\pi \Delta)} < \frac{1}{2}$$

$$E[f(y)] \neq f(E(y))$$

Q2 of A2:

$$Q2 | z_1, \dots, z_n \sim N(\mu, \sigma^2).$$

$$y_i = f(\text{out } z_i)$$



$$y_i \rightarrow 0 \quad 1 \quad 2 \quad 3 \quad 4$$

$$Q^L(\mu, \sigma^2 | y_i, z_i, i=1, \dots, n)$$

$$= \sum_{i=1}^n \log f(y_i, z_i | \mu, \sigma^2)$$

$$= \sum_{i=1}^n \log \underbrace{f(z_i | \mu, \sigma^2) \cdot f(y_i | z_i)}$$

$$= \sum_{i=1}^n \log \frac{1}{\sqrt{\sigma^2}} e^{-\frac{(z_i - \mu)^2}{2\sigma^2}} \cdot f(y_i | z_i) \quad + \textcircled{C}$$

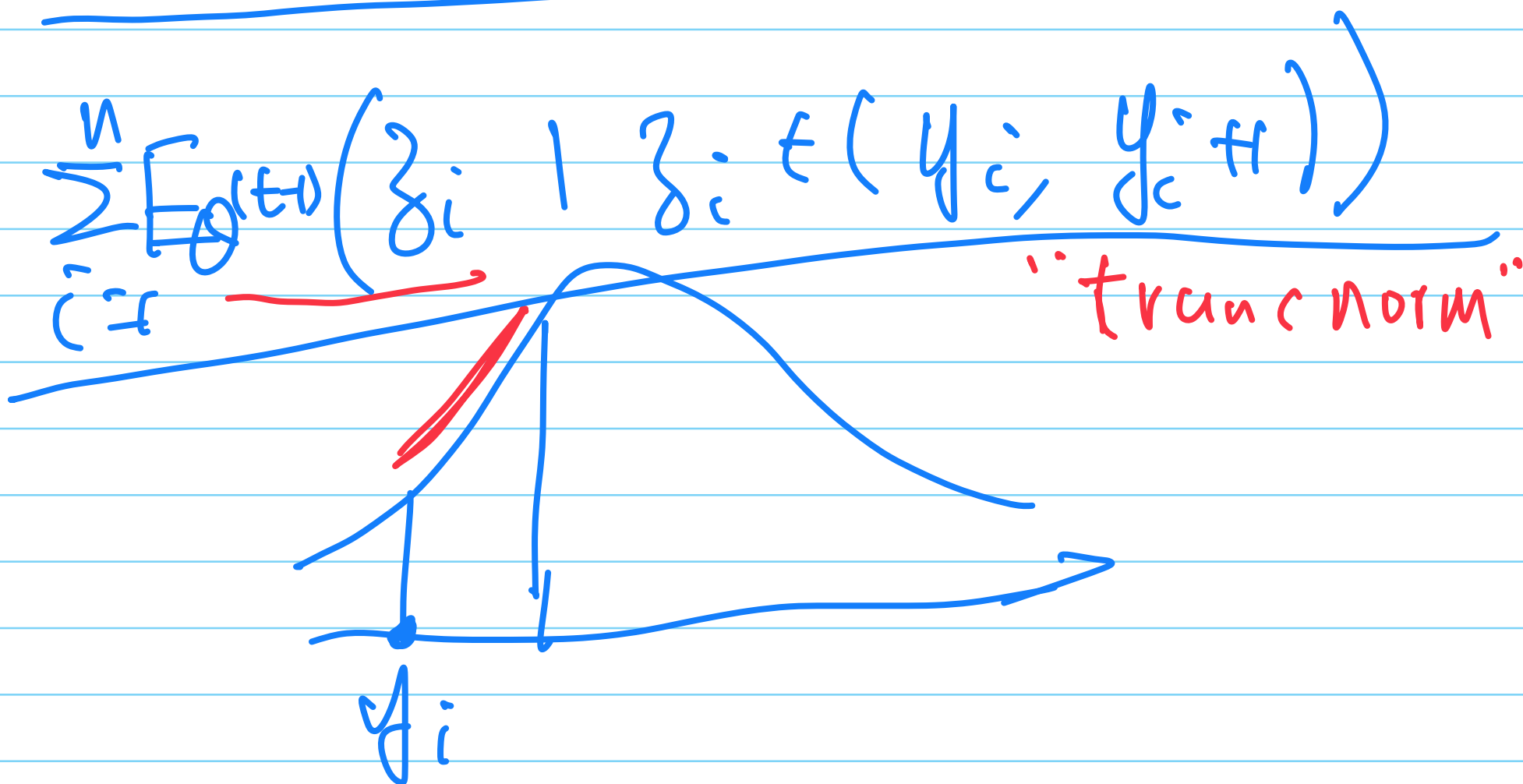
$$= \sum_{i=1}^n \left[-\frac{1}{2} \log \sigma^2 - \frac{(z_i - \mu)^2}{2\sigma^2} \right]$$

$$= -\frac{n}{2} \log \sigma^2 - \frac{1}{2} \frac{\sum_{i=1}^n (z_i - \mu)^2}{\sigma^2}$$

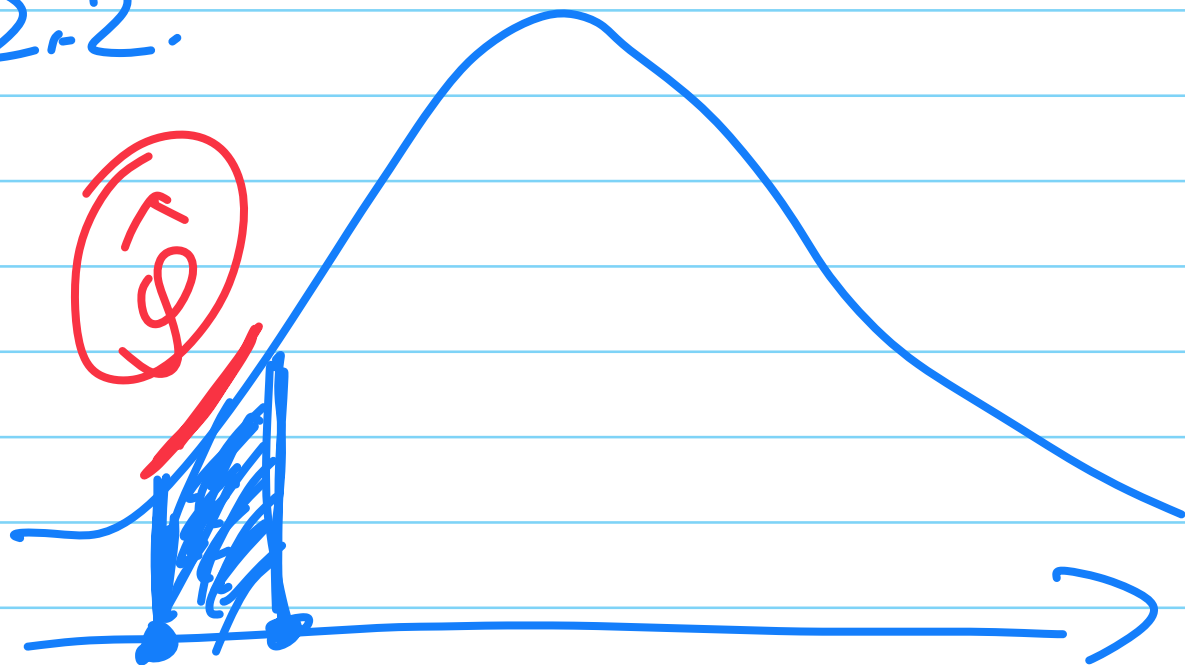
$$= -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (z_i^2 - 2\mu z_i + \mu^2)$$

Γ -step

$$\sum_{i=1}^n F_0^{(t+1)}(z_i \mid z_i \in (y_i, y_{i+1}))$$



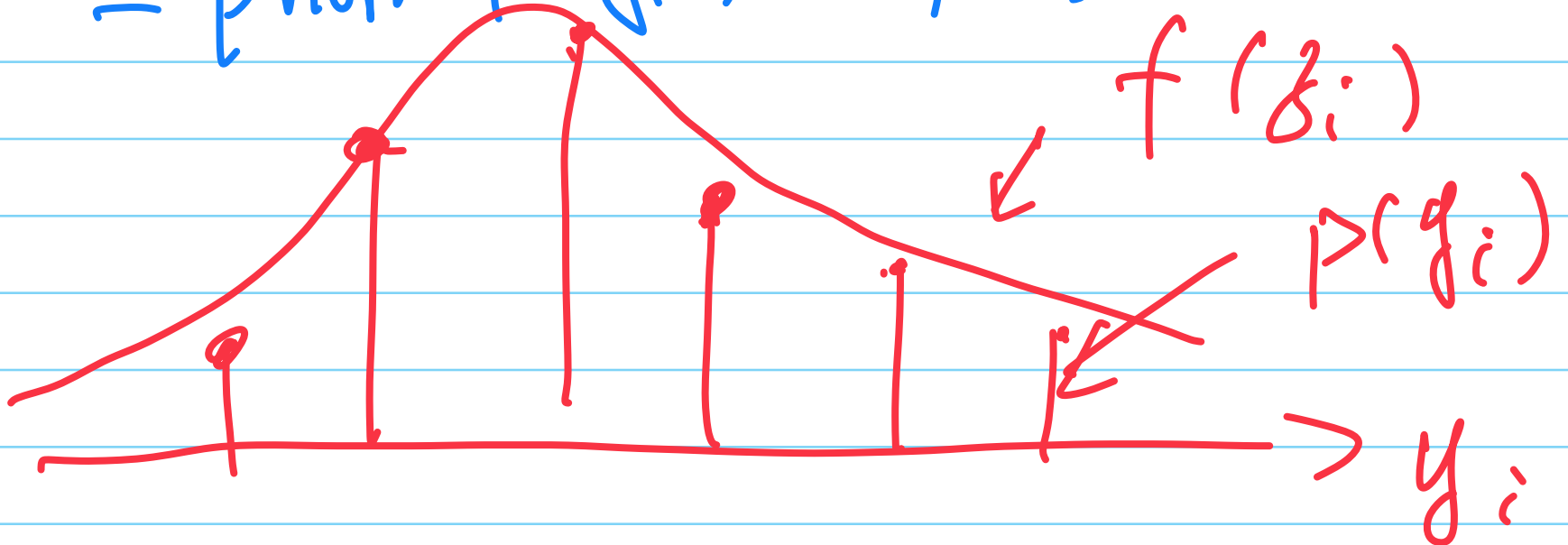
Q2.2.



$$P(y_i | \mu, \sigma^2) = \int_{y_i}^{y_{i+1}} \phi(z_i | \mu, \sigma^2) dz_i$$

$$= \text{pnorm}(y_{i+1}, \mu, \sigma)$$

$$- \text{pnorm}(y_i, \mu, \sigma)$$



Example

$$Z_1, \dots, Z_n \sim N(\mu, \sigma^2)$$

$$Z_1, \dots, Z_m, Z_{m+1} < 0, \dots, Z_n < 0$$

> 0

How to use EM algorithm
to est. μ & σ^2 ?